

1. Given a list of integers, return the largest product that can be made by multiplying any three integers.

For example, if the list is `[-10, -10, 5, 2]`, we should return 500, since that's $-10 * -10 * 5$

2. A number is considered perfect if its digits sum up to exactly 10.

Given a positive integer `n`, return the `n`-th perfect number.

For example, given 1, you should return 19. Given 2, you should return 28.

3. Given an array of numbers, find the length of the longest increasing subsequence in the array. The subsequence does not necessarily have to be contiguous.

For example, given the array `[0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15]`, the longest increasing subsequence has length 6: it is 0, 2, 6, 9, 11, 15.

4. You are given an `N` by `M` 2D matrix of lowercase letters. Determine the minimum number of columns that can be removed to ensure that each row is ordered from top to bottom lexicographically. That is, the letter at each column is lexicographically later as you go down each row. It does not matter whether each row itself is ordered lexicographically.

For example, given the following table:

`cba`

`daf`

`ghi`

This is not ordered because of the `a` in the center. We can remove the second column to make it ordered:

`ca`

`df`

`gi`

So your function should return 1, since we only needed to remove 1 column.

As another example, given the following table:

`abcdef`

Your function should return 0, since the rows are already ordered (there's only one row).

As another example, given the following table:

`zyx`

`wvu`

`tsr`

Your function should return 3, since we would need to remove all the columns to order it.

5. Given a list of possibly overlapping intervals, return a new list of intervals where all overlapping intervals have been merged. The input list is not necessarily ordered in any way. For example, given [(1, 3), (5, 8), (4, 10), (20, 25)], you should return [(1, 3), (4, 10), (20, 25)].